

Solution Architecture

Auto Ticket Classification using Flow Designer

Date	2 oct 2026
Team ID	SWTID-2026-5846
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	4 Marks

1. Solution Architecture

The solution architecture describes how the business requirement for automatic IT ticket classification is translated into a ServiceNow implementation. The architecture separates the user-facing ticket form, ticket data storage, workflow logic, classification rules and notification step while keeping these components within the ServiceNow environment.

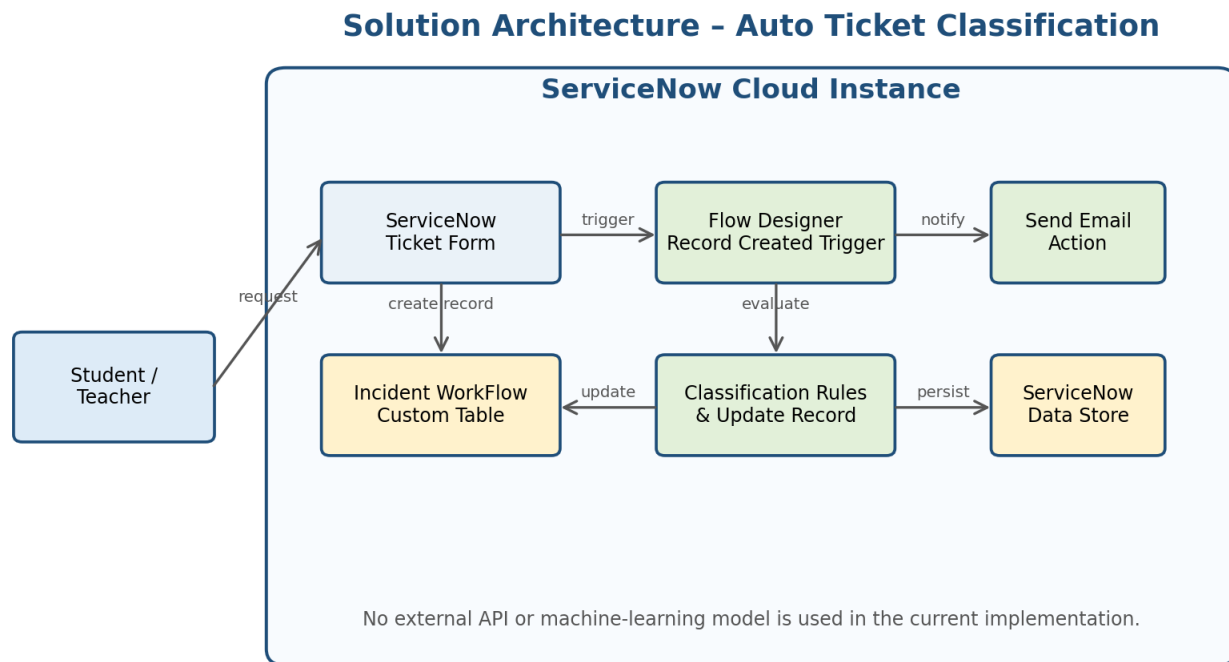


Figure 1. Architecture and data flow of the proposed solution.

2. Architecture Components

Component	Responsibility
Student / Teacher	Creates the IT support request and provides caller information and a short description of the issue.
ServiceNow Ticket Form	Provides the interface through which the support request is entered.

Incident WorkFlow Custom Table	Stores the ticket record and fields such as Number, Caller, Category, Subcategory, Short Description, Description, State, Assigned Group and Assigned To.
Flow Designer	Starts when a new record is created and controls the automatic classification workflow.
Classification Rules	Checks the Short Description against predefined issue keywords and assigns the corresponding Category and Subcategory.
Choice / Dependency Configuration	Maintains valid Category and Subcategory relationships.
Update Record	Writes the classification result back to the current ticket record.
Send Email Action	Sends the confirmation message to the caller.
ServiceNow Data Store	Maintains the configured ticket records within the ServiceNow platform.

3. End-to-End Processing Flow

1. A student or teacher opens the ServiceNow ticket form.
2. The user enters the caller information and Short Description and submits the request.
3. ServiceNow creates a record in the Incident WorkFlow custom table.
4. The Record Created trigger starts the Auto Classify School IT Tickets flow.
5. The flow evaluates the Short Description using the configured classification conditions.
6. The matching condition updates the Category and Subcategory fields.
7. The updated classification is stored in the ticket record.
8. The Send Email action sends a confirmation to the caller.

4. Classification Layer

Issue / Keyword	Category	Subcategory
WiFi / Wi-Fi	Network	Wi-Fi
Projector	Hardware	Projector
Password / Login	Access	Forgot Password
Slow / Hanging	Performance	Slow Computer

5. Data and Configuration Design

The Incident WorkFlow table acts as the central record structure for the project. The table uses reference fields for user-related information, choice fields for controlled values, and text fields for the issue description. The Subcategory field is configured as dependent on Category so that the classification relationship remains consistent.

Field	Type	Purpose
Number	Auto Number	Unique ticket identifier.
Caller	Reference – sys_user	Identifies the person raising the request.
Category	Choice	Stores Network, Hardware, Access or Performance.
Subcategory	Choice / Dependent	Stores the issue subtype associated with the Category.
Short Description	String	Primary text evaluated by the classification flow.

Description	String	Provides additional issue details.
State	Choice	Tracks the ticket lifecycle state.
Assigned Group	Reference – sys_user_group	Identifies the support group.
Assigned To	Reference – sys_user	Identifies the assigned support user.

6. Architecture Characteristics

- Platform: ServiceNow cloud environment.
- Automation: ServiceNow Flow Designer.
- Data storage: Incident WorkFlow custom table within ServiceNow.
- External interfaces: No external API is required by the current implementation.
- Machine learning: Not used; classification is based on predefined keyword rules.
- Notification: ServiceNow Flow Designer Send Email action.
- Configuration management: Project Update Set.

7. Architecture Boundary

The current architecture is intentionally limited to the components required for the academic project. It does not include an external application server, separate database, third-party API or machine-learning service. Future enhancements can introduce additional components only when they are required and implemented.

8. Stakeholder View

From the user perspective, the system remains simple: provide the issue description and submit the request. From the IT helpdesk perspective, the system provides a structured record and automatic classification. From the administrator perspective, the classification rules and table configuration can be maintained through ServiceNow configuration.